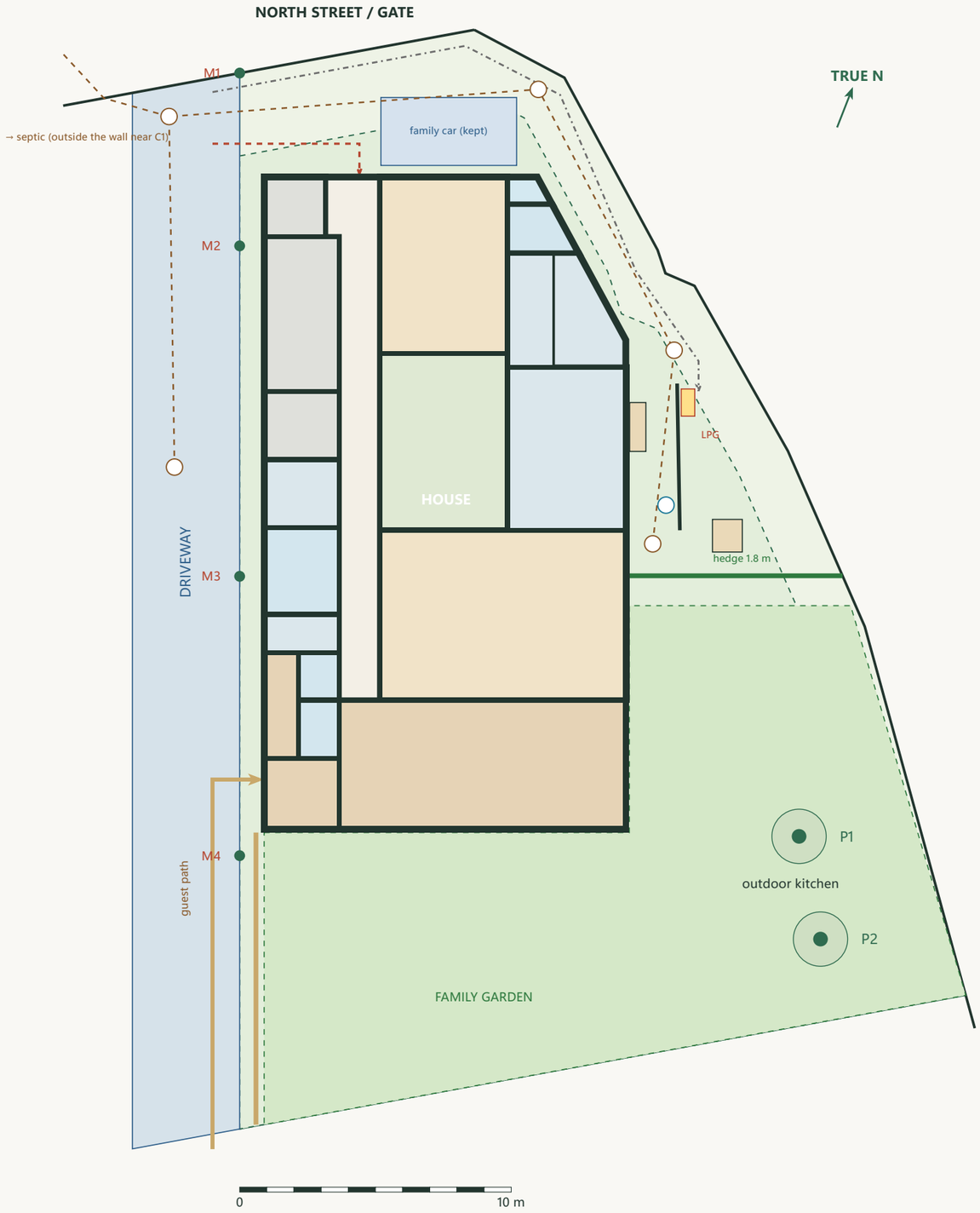
The house on its site · the site and services plan

Site option Z4 | shift (0.91, 3.825) m, rotation 0° | the first floor stands on the same outline | 250 mm outer / 200 mm inner walls | V8.2: the site plan (`studies/site_plan.typ`, every position proposed, to confirm with the survey and the engineers)



318.77 m² outside footprint (incl. courtyard)	**277.76 m²** ground floor rooms (axes)	**220.43 m²** first floor rooms (axes)
4.5 × 6.3 m courtyard, clear · sky 3.15 m axes above	**55.6 m²** master suite (indoors)	**0.785 m** west face to the M line (both floors)
0.2 m nearest wall face to the envelope	**5.99 / 7.84 m** palms P1 / P2 to the outer walls	**3.6 m** floor to floor (to confirm)

The house. A compact courtyard house on site option Z4: a pentagon 13.311 × 24 m on wall centre lines, its north-east corner cut parallel to the lane, square to the M line and pushed as far north as the setbacks allow. The courtyard is the cool heart; the rooms look in, not out. A west service band and an east service block frame the middle band (women's majlis, courtyard, double-height living); the men's majlis takes the full width on the south.

First floor. The family part on the north and west around the courtyard; the south wing (V8.0): the gym on the west, a large suite court open to the sky, the owner's office on the east (reached from the court only), and the master suite. The gallery runs on a 1.55 m projection over the courtyard and the double-height void, on an edge beam and 3 columns.

Roof access (V8.1): a maintenance ladder in a recess of the gym's east wall in the suite court, behind a locked walnut panel; the internal stair stops at the first floor; office visitors are met in the men's majlis. One entry to the site: the farm's 4 m main gate.

Walls on walls: the first-floor outer walls stand on the ground-floor outer walls, 0.785 m from the M line; no projection beyond the ground floor.

Site and services plan (V8.2, a study; every position proposed, to confirm with the survey and the engineers): one entry, the farm's 4 m gate; the family's covered parking beyond the plot and one uncovered car in the north strip (kept); guests walk the honey-stone **guest path** along the driveway to their entrance; workers reach the **service yard** (bins, screen, LPG cabinet) along the north and lane-side strips; the **family garden on the south only**, behind a 1.2 m stone wall and a 1.8 m hedge. Foul drains to the existing **septic outside the wall near corner C1**; the water tank east of peg R with the reserved tank beside it; rainwater separate; PV provision with the inverter and batteries in the utility room, no generator; intercom and seven cameras. The plumbing study (rounds 1–2) is a proposal.

المنزل

منزل فناء مدمج على خيار الموقع Z4: مخمس 13.311 × 24 م على محاور الجدران، ركنه الشمالي الشرقي مقطوع موازيا للزقاق، متعامد مع خط M ومدفوع شمالا قدر ما تسمح الارتدادات. الفناء هو القلب البارد، والغرف تطل إلى الداخل لا إلى الخارج.

الدور الأول

جزء العائلة شمالا وغربا حول الفناء، والجناح الجنوبي (V8.0): النادي غربا وفناء الجناح الكبير المفتوح للسماء ومكتب المالك شرقا، ثم جناح النوم الرئيسي. الممر العلوي على بروز 1.55 م فوق الفناء والصالة، على جسر حافة و3 أعمدة.

الوصول إلى السطح (V8.1)

سلم صيانة في تجويف بجدار النادي داخل فناء الجناح خلف لوح خشبي مقفل؛ الدرج الداخلي يتوقف عند الدور الأول؛ زوار المكتب يُستقبلون في مجلس الرجال. مدخل واحد للموقع: البوابة الرئيسية للمزرعة بعرض 4 م.

جدار فوق جدار

الجدران الخارجية للدور الأول فوق جدران الأرضي، على بعد 0.785 م من خط M، ولا بروز خارج الأرضي.

مخطط الموقع والخدمات (V8.2، دراسة)

مدخل واحد (بوابة المزرعة 4 م)؛ موقف سيارة مكشوف في الشريط الشمالي؛ ممر حجري للضيوف بمحاذاة الممر الغربي؛ ساحة الخدمة شرقا؛ حديقة العائلة جنوبا فقط خلف جدار حجري وسياج؛ الصرف إلى خزان التحليل القائم خارج السور قرب C1؛ خزان الماء شرق الوتد R؛ الأمطار منفصلة؛ تجهيز شمسي بلا مولد. كل موقع مقترح حتى الرفع المساحي.

Ground floor

Room numbers G.. | dark squares = the 3 columns, orange dashed = the first-floor edge beam above (a colonnade along the courtyard) | doors and windows outside the two studied blocks are indicative



Middle band: the women's majlis 4.7 × 6.5 m, the courtyard 4.7 × 6.5 m (4.5 × 6.3 m clear, open to the sky) and the living room in two heights: double height on the courtyard, normal height with the dining to the east. The living is entered from the riwaq through a 1.4 m arched opening; its west wall is solid and full height with the arched TV niche (V6.3).

West band: the lift (installed at construction, V6.3), the open stair (flipped, V8.2: the lower flight rises south off the lobby behind a black-bar balustrade; a full-height coat cupboard in the west half of its start strip, the store under the half landing at the south end), the utility room with the shaft, then the west wash block (sheet 06). The riwaq 2 × 19.25 m runs along the courtyard.

East block (sheet 06): kitchen, laundry, maid room and maid bath, the air tower in the cut corner. **Men's majlis** 10.56 × 4.75 m, entered only from the guest foyer. No east duct. **Colonnade:** 3 columns on x 5.8 under the first-floor edge beam, in the women's majlis south wall, the courtyard / living glass and the living's south wall, with a three-arch arcade in the courtyard bay (V7.1).

Courtyard (revised V8.3): the whole floor level (no step), modern movable sofas and armchairs for 4–6 with a low table facing the outdoor TV in its semicircular honey-stone surround on the kitchen wall (the malqaf grille below it); the water + tree enclosure on the kitchen wall in front of the sink window (y 7.3–8.9): the tree north, a raised water trough south, the view from the sink; gentle falls to linear drains on the rain system; the north side in front of the women's majlis glass free; no pergola. **Outside (V8.3):** the outdoor counter and sink next to the kitchen service door (canopy optional), the grill and tannour a little apart in the yard's south part (≥ 3 m from every opening, downwind); the yard screen, the gas cabinet at its north end, the bins north of the door.

الوسط: مجلس النساء والفناء والصالة

الفناء 4.7 × 6.5 م على المحاور و4.5 × 6.3 م صافيا مفتوح للسماء. الصالة بسقفين: ارتفاع مزدوج على الفناء وارتفاع عادي مع الطعام شرقا، وتُدخل من الرواق عبر فتحة مقوسة 1.4 م؛ جدارها الغربي مصمت كامل الارتفاع مع كوة التلفاز المقوسة (V6.3).

الشريط الغربي والرواق

المصعد والدرج المفتوح (معكوس الاتجاه في V8.2: الجزء السفلي يصعد جنوبا من الردهة خلف درابزين أسود، ودولاب المعاطف في النصف الغربي من شريط البداية) وغرفة الخدمات مع الشفت ثم كتلة المغاسل (اللوحة 06). الرواق 2 × 19.25 م على طول الفناء.

الكتلة الشرقية ومجلس الرجال

المطبخ والغسيل وغرفة الخادمة وحمامها وبرج الهواء (اللوحة 06). مجلس الرجال 10.56 × 4.75 م لا يدخل إلا من بهو الضيوف. لا شفت شرقي.

الأعمدة والرواق

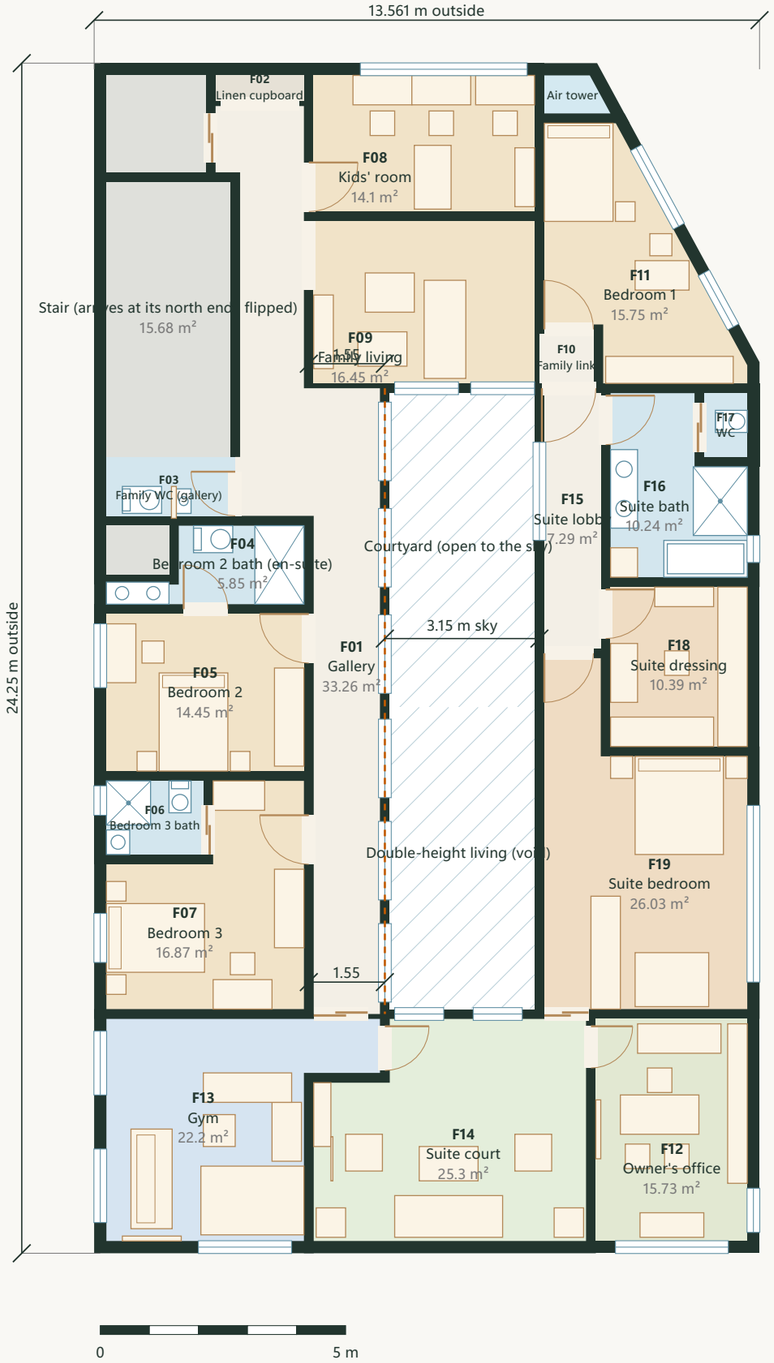
3 أعمدة على 5.8 x (في جدار مجلس النساء الجنوبي، وبين زجاج الفناء والصالة، وفي جدار الصالة الجنوبي) تحمل جسر حافة البروز في الدور الأول (الخط البرتقالي المتقطع)، فيصبح على حافة الفناء رواق بأعمدة وثلاثة أقواس.

الفناء (معدل V8.3)

أرضية مستوية بلا درجة، أثاث متحرك عصري لأربعة إلى ستة مع طاولة منخفضة أمام التلفاز الخارجي في إطاره الحجري على جدار المطبخ وتحته فتحة الملقف، والنافورة والشجرة على جدار المطبخ أمام نافذة المغسلة، ومصارف خطية إلى نظام الأمطار؛ والمطبخ الخارجي بجوار باب الخدمة والشواية على بعد ≥ 3 م من الفتحات. وض ماء وشجرة في الطرف الشمالي، بلا عريشة. نافذة حوض المطبخ عند 7.3–8.9 y.

First floor

Room numbers F.. | west projection 1.55 m over the courtyard and the void, edge beam (orange dashed) on 3 columns | hatched = open below / to the sky | furniture from the two first-floor studies



TRUE N

West projection: 1.55 m over the courtyard and the double-height void, on an edge beam on x 5.8 and 3 columns; the first-floor west wall stands on the GF west wall. Gallery 1.55 m (≈ 1.42 m clear), west rooms 4.25 m (≈ 4.025 m clear), courtyard sky 3.15 m (≈ 2.95 m clear).

North band: kids' room 14.1 m² and family living 16.45 m² on the courtyard edge; bedroom 1 15.75 m².

West band: family bath, bedroom 2, bedroom 3 with its own bath; every door from the gallery.

South wing v2 (V8.0): the gym on the west band 22.2 m² (the treadmill with its manufacturer zone), the suite court 25.3 m² open to the sky from the gym to the office, the owner's office 15.73 m² on the east (no door to the bedroom; reached from the court only); master suite 55.6 m².

البروز الغربي والممر العلوي

بروز 1.55 م فوق الفناء وفراغ الصالة على جسر حافة على 5.8 x و3 أعمدة، والجدار الغربي للدور الأول فوق الجدار الغربي للأرضي. الممر 1.55 م (نحو 1.42 م صافي)، والغرف 4.25 م (نحو 4.025 م صافي)، والسماء فوق الفناء 3.15 م.

الشريط الشمالي

غرفة الأطفال 14.1 م² وجلسة العائلة 16.45 م² على حافة الفناء، وغرفة النوم 1 15.75 م².

الشريط الغربي

حمام العائلة بجوار الشفت، وغرفة النوم 2، وغرفة النوم 3 مع حمامها فوق دورة العائلة؛ كل الأبواب من الممر ولا باب على بسطة الدرج.

الجنوب والشرق

النادي غربا 22.2 م²، وفناء الجناح 25.3 م² المفتوح للسماء من النادي إلى المكتب، ومكتب المالك شرقا 15.73 م² بلا باب إلى غرفة النوم (يُدخل إليه من فناء الجناح فقط)، وجناح النوم الرئيسي 55.6 م² بمدخل من وصلة العائلة.

First floor / the family part

Kids' room, family living, bedrooms 1–3, family bath, gallery on the west projection | furnished test from studies/ff_family.typ



Kids' room 4.7 × 3 m: two study desks and a PC desk under the north window, a gaming corner; solid door from the gallery. **Family living** 4.7 × 3.5 m on the courtyard edge: a view, not a path.

Bedroom 2 4.25 × 3.4 m with its own bath beside the shaft (3 Nov 2026; the family WC next to it, from the gallery); **bedroom 3** 16.87 m² with its own bath 2.2 × 1.7 m over the GF family WC; **bedroom 1** 15.75 m², one room without a bath.

Bedroom 2's bath 4.25 × 1.8 m and the **family WC** 2.75 × 1.3 m by the west shaft (3 Nov 2026: the F-b bath split; every fixture on one service wall into the shaft), the family WC's door from the gallery, for bedroom 1, the family living and the kids' room.

غرفة الأطفال: دراسة وألعاب

4.7 × 3 م: مكتبا دراسة ومكتب كمبيوتر تحت النافذة الشمالية وركن ألعاب، وباب مصمت من الممر. يمكن أن تصبح غرفة نوم رابعة لاحقا.

جلسة العائلة

4.7 × 3.5 م = 16.45 م² على حافة الفناء: إطلالة لا ممر، والطريق إلى جناح الوالدين خلف الكنبة.

غرف النوم

النوم 2: 4.25 × 3.4 م بتجويف دولاب بجانب الشفت. النوم 3: 16.87 م² مع حمام خاص 2.2 × 1.7 م فوق دورة العائلة. النوم 1: 15.75 م² غرفة واحدة بلا حمام.

حمام العائلة

حمام غرفة النوم 2 4.25 × 1.8 م ودورة المياه العائلية 2.75 × 1.3 م بجوار الشفت الغربي (3 نوفمبر 2026)، وباب دورة المياه من الممر.

التخزين

دولاب البياضات في نهاية الممر الشمالية (2 × 0.7 م). الدواليب المدمجة صناديق نجارة مثبتة بين الجدارين.

First floor / the south wing and the master suite

South wing v2 (owner, V8.0): the gym on the west band, the suite court from the gym to the office, the office on the east | furnished test from studies/ff_suite.typ | orange = the treadmill's clear zones, red = the double wall to bedroom 3 | the three court doors on a 1.2 m strip



Master suite 55.6 m² (unchanged): a lobby from the family link opens to the bath, the dressing and the bedroom (the bed is not seen from the suite door). Luxury bath 11.89 m²; walk-in dressing 3.01 × 3.45 m; bedroom 26.03 m². **No door to the office.** Its court door (sliding glass) opens onto an entry zone formed by a full-height wardrobe: the sight-line test (doors open, from the gym and office doors and from the court seating) keeps the bed out of view.

Suite court 25.3 m² on axes (≈ 23.1 m² clear), open to the sky, from the gym to the office: the three court doors (bedroom, gym, office) on a 1.2 m strip along its north edge, the seating (a sofa, two armchairs, a table, plants) in the south part; the floor falls south to a box gutter with an overflow, linear drains at the near-flush doors. Its south side is the facade frame: a 1.10 m base under R8, the cornice and the parapet; the infill is the owner's choice (a mashrabiya screen recommended). The clerestory windows of the living get the winter noon sun on 100 % of their height.

Gym 22.2 m² on the west band (inside the column frame), with an entrance nib at the gallery end: the treadmill along the length on the stiff 2.75 m bay (0.5 m clear each side, 2 m behind), a power rack, kettlebells, a bike, the pilates mat, a fixed TV; windows west and south; a double wall to bedroom 3 (Rw ≈ 55 dB).

Owner's office 3.31 × 4.75 m (east): for the owner only, reached from the suite court (bedroom → court → office; the family gym → court → office); visitors are met in the men's majlis; a simple door, an east window, a fixed TV.

Roof ladder (V8.1): a manufactured vertical ladder in a 0.90 × 0.30 recess of a honey-stone pilaster on the gym's east wall (north end, off the door strip and the seating), behind a locked walnut-finish panel; 3.70 m to the gym roof, the stiles 1.10 m above it, a self-closing gate in the court guard (`studies/roof.typ` page 2).

جناح النوم الرئيسي · 55.6 م²

كما هو: ردهة من وصلة العائلة إلى الحمام والملابس والنوم. لا باب بين غرفة النوم والمكتب. باب الفناء الزجاجي المنزلق يفتح على منطقة دخول يكوّنها دولاب كامل الارتفاع، فلا يُرى السرير من باب النادي أو المكتب أو من جلسة الفناء والأبواب مفتوحة.

فناء الجناح

25.3 م² مفتوح للسماء من النادي إلى المكتب: الأبواب الثلاثة على ممر بعرض 1.2 م في طرفه الشمالي والجلسة في الجنوب، والأرضية تميل جنوبا إلى مزراب مع فائض، ومصارف خطية أمام الأبواب شبه المستوية. جهته الجنوبية إطار الواجهة بجدار سفلي 1.10 م، والحشوة اختيار المالك (المشربية مقترحة).

النادي

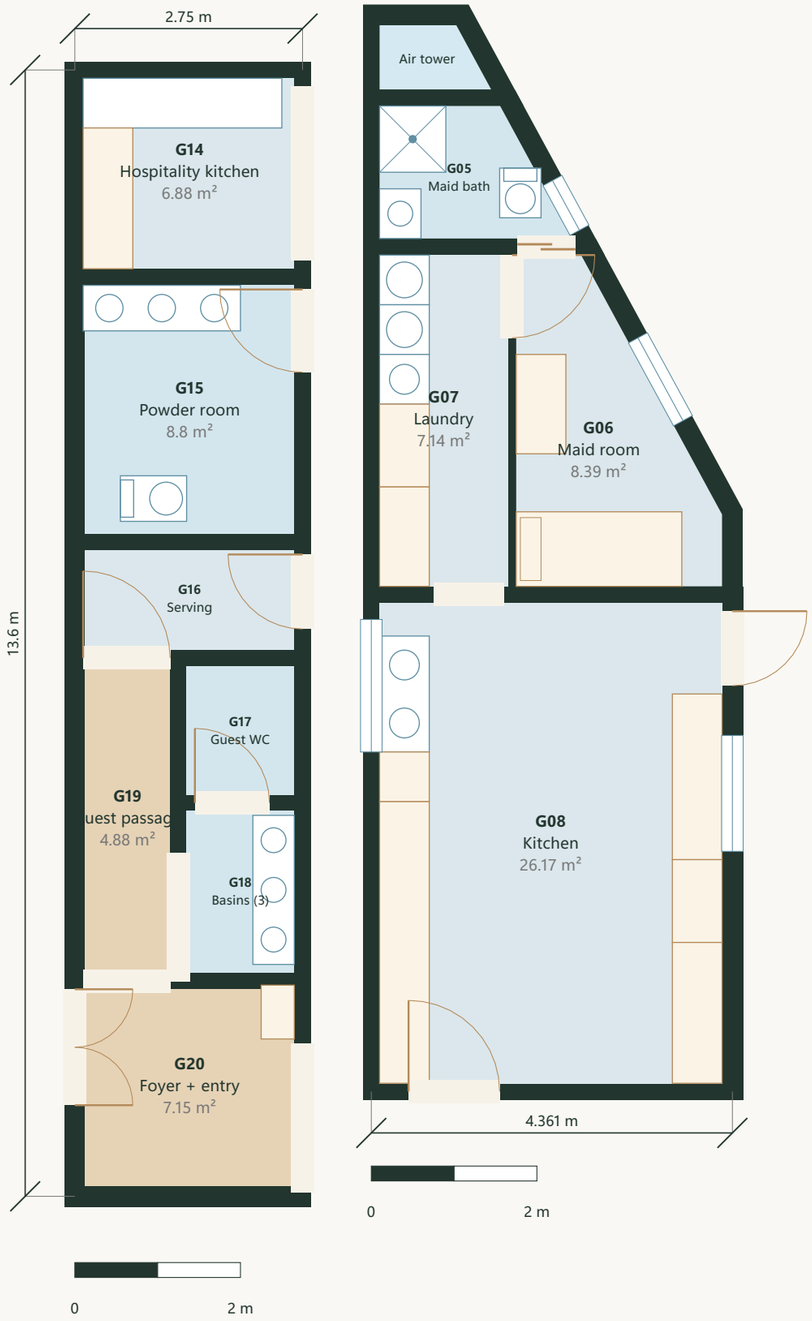
22.2 م² في الشريط الغربي ضمن إطار الأعمدة، بمدخل من نهاية الممر: جهاز المشي بطول الغرفة على البلاطة الصلبة، ورف الأثقال والكيتل بل والدراجة وبساط البيلاتس وتلفاز ثابت؛ جدار مزدوج مع غرفة النوم 3.

مكتب المالك

شرقا للمالك فقط، يُدخل إليه من فناء الجناح؛ الزوار في مجلس الرجال؛ باب بسيط ونافذة شرقية وتلفاز ثابت. سلم الصيانة إلى السطح في تجويف بجدار النادي خلف لوح خشبي مقفل.

Ground floor / the west wash block and the east service block

Room studies studies/guest_wing.typ and studies/east_wing.typ | doors, fixtures and walls as studied



West wash block (2.75 × 13.6 m): one strip of wet rooms on one drain run. Family side, open to the riwaq: hospitality kitchen 2.75 × 2.5 m (a light kitchen for coffee, tea, dates and desserts); the family powder room 2.75 × 3.2 m (3 Nov 2026: the basins and the WC as one luxurious room with a door from the riwaq; no shower on the ground floor).

The **serving passage** 2.75 × 1.4 m (2.53 × 1.2 m clear) is the only family / guest link: door 1 (0.9 m) from the riwaq and door 2 (1.05 m) from the guest side are never open together and both lock from the family side.

Guest side: guest WC behind a screen wall, one trough with three taps, the guest passage and the foyer with a 1.4 m double door and a 1.8 m opening into the men's majlis (a bent entry, partial).

East service block: kitchen 4.36 × 6 m on the courtyard lines (two parallel runs, no island, door to the dining only); the sink under its courtyard window (y 7.3–8.9, V8.0) with the dishwasher south of it, the outdoor TV on the other face of the wall further south; laundry 1.7 × 4.2 m on the solid wall of the women's majlis; maid room 8.39 m² (an escape casement, sill +1.10) and maid bath 3.84 m² (the WC in the cut-wall corner, for the air channel's throat) in the sloped corner, reached only through the kitchen and the laundry; the air tower in the tip of the cut corner: one tower, two separate paths (the courtyard breeze in a shallow channel in the slab; the HRV's intake, the unit in its base on the roof).

كتلة المغاسل الغربية

خط واحد من الغرف الرطبة على خط صرف واحد. جهة العائلة مفتوحة على الرواق: مطبخ الضيافة 2.75 × 2.5 م للقهوة والشاي والتمر والحلى، ومغاسل العائلة، ودورة العائلة.

ممر التقديم

2.75 × 1.4 م (2.53 × 1.2 م صافي) هو الصلة الوحيدة بين العائلة والضيوف: الباب 1 (0.9 م) من الرواق والباب 2 (1.05 م) من جهة الضيوف لا يفتحان معا ويقفلان من جهة العائلة.

جهة الضيوف

دورة الضيوف خلف جدار ساتر، ومغسلة حوضية بثلاثة صنابير، وممر الضيوف، والبهو بباب مزدوج 1.4 م وفتحة 1.8 م إلى المجلس (مدخل منكسر جزئي).

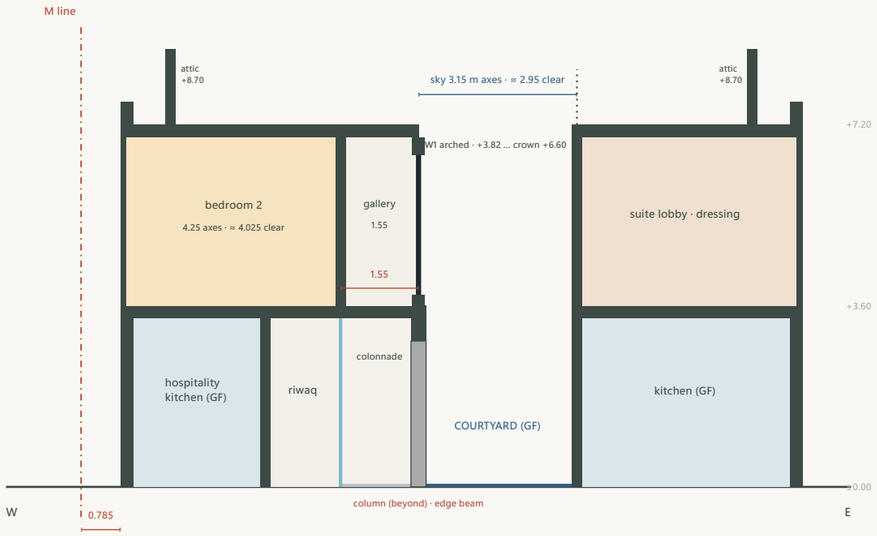
الكتلة الشرقية

المطبخ 4.36 × 6 م على خطوط الفناء بخطي خزائن وبلا جزيرة وبابه إلى الطعام فقط، والحوض تحت نافذته على الفناء (y 7.3–8.9) وغسالة الصحون جنوبه؛ والغسيل 1.7 × 4.2 م على جدار مجلس النساء المصمت، وغرفة الخادمة 8.39 م² بنافذة هروب وحمامها 3.84 م² في الركن المائل؛ وبرج الهواء في رأس الركن المقطوع: برج واحد بمسارين منفصلين (نسيم الفناء ومدخل هواء وحدة التهوية في قاعدته على السطح).

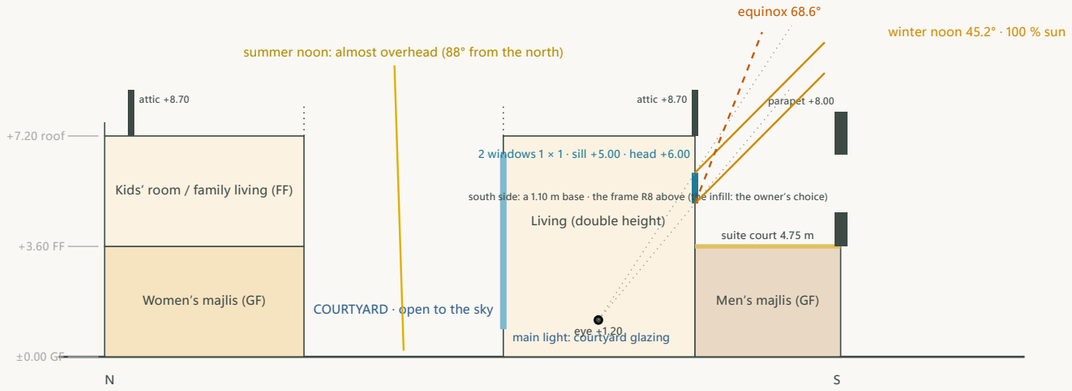
Sections / the west projection and the light

Schematic sections, levels to confirm: GF ±0.00 · FF +3.60 · roof +7.20 | the column on x 5.8 is beyond the cut | the light figures are geometry only

A · east–west through the courtyard (y 12.0), looking north



B · north–south through the middle band (x ≈ 7.3), looking west · the light check



Behind the 4.75 m suite court	Winter noon sun on the windows	Equinox noon sun	Sky seen from the eye
the south facade frame with its parapet (top +8.00) · **V8.0, south wing v2**	100 %	100 %	100 %
the same frame without a parapet (solid top +7.20)	100 %	100 %	100 %

A · west projection: the gallery stands on a 1.55 m slab over the courtyard, carried by an edge beam on x 5.8 (spans 6.5 and 6.25 m) on 3 columns: on the ground floor a colonnade along the courtyard, with a normal-height ceiling under the projection. The first-floor west wall stands on the ground-floor west wall, 0.785 m from the M line. Courtyard sky 3.15 m on axes (≈ 2.95 m clear).

B · light: the main light of the double-height living comes from the courtyard; its 2 clerestory windows face the suite court, which since V8.0 (south wing v2) runs 4.75 m to the south facade. Over the facade's parapet the winter noon sun reaches 100 % of the window height and the sky shows on 100 % (V7.1, behind the 2.0 m court: 81 %).

المقطع A · البروز الغربي

يقوم الممر على بلاطة 1.55 م فوق الفناء يحملها جسر حافة على 5.8 x فوق 3 أعمدة، فيصبح في الأرضي رواق بأعمدة على حافة الفناء بسقف عادي تحت البروز. الجدار الغربي للدور الأول فوق الجدار الغربي للأرضي على بعد 0.785 م من خط M. السماء فوق الفناء 3.15 م على المحاور.

المقطع B · الضوء

الضوء الرئيسي للصالة من زجاج الفناء. النوافذ العلوية (2) تطل على فناء الجناح الممتد 4.75 م حتى الواجهة الجنوبية (V8.0)، فتصل شمس ظهر الشتاء إلى 100 % من ارتفاع النافذة.

الخصوصية

جلسة النوافذ فوق مستوى النظر من أرضية فناء الجناح أو زجاج معتم.

Areas / room schedules

Axes = wall centre lines | clear = inside the walls as drawn (250 mm outer, 200 mm inner, 100 mm partitions, the steel windows on the courtyard edges), furniture and thresholds not deducted

Ground floor

No.	Space	Axis size	Axis area	≈ Clear
G01	Courtyard	4.7 × 6.5 m	30.55 m²	–
G02	Women's majlis	4.7 × 6.5 m	30.55 m²	28.24 m²
G03	Living (double height)	4.7 × 6.25 m	29.38 m²	27.83 m²
G04	Men's majlis	10.56 × 4.75 m	50.16 m²	46.77 m²
G05	Maid bath	polygon	3.84 m²	3.03 m²
G06	Maid room	polygon	8.39 m²	7.28 m²
G07	Laundry	1.7 × 4.2 m	7.14 m²	6.2 m²
G08	Kitchen	4.36 × 6 m	26.17 m²	23.99 m²
G09	Living (normal height) · dining	4.36 × 6.25 m	27.26 m²	25.63 m²
G10	Lift (shaft placeholder)	2.25 × 2.2 m	4.95 m²	4 m²
G11	Family lobby + riwaq	polygon	29.97 m²	25.72 m²
G12	Stair	2.75 × 5.7 m	15.68 m²	13.89 m²
G13	Utility room (shaft inside)	2.75 × 2.5 m	6.88 m²	5.81 m²
G14	Hospitality kitchen	2.75 × 2.5 m	6.88 m²	5.81 m²
G15	Family powder room (WC · 2 basins)	2.75 × 3.2 m	8.8 m²	7.58 m²
G16	Serving passage (2 doors)	2.75 × 1.4 m	3.85 m²	3.03 m²
G17	Guest WC	1.5 × 1.75 m	2.63 m²	2.02 m²
G18	Guest basins (trough · 3 taps)	1.5 × 2.15 m	3.22 m²	2.53 m²
G19	Guest passage	1.25 × 3.9 m	4.88 m²	3.79 m²
G20	Foyer + entry	2.75 × 2.6 m	7.15 m²	6 m²

First floor

No.	Space	Axis size	Axis area	≈ Clear
F01	Upper gallery (on the west projection)	polygon	33.26 m²	29.38 m²
F02	Linen cupboard	2 × 0.7 m	1.4 m²	0.95 m²
F03	Family WC (gallery)	2.75 × 1.3 m	3.57 m²	3.15 m²
F04	Bedroom 2 bath (en-suite)	polygon	5.85 m²	4.67 m²
F05	Bedroom 2	4.25 × 3.4 m	14.45 m²	12.88 m²
F06	Bedroom 3 bath	2.2 × 1.7 m	3.74 m²	2.96 m²
F07	Bedroom 3	polygon	16.87 m²	15.02 m²
F08	Kids' room (study + gaming)	4.7 × 3 m	14.1 m²	12.49 m²
F09	Family living	4.7 × 3.5 m	16.45 m²	14.85 m²
F10	Family link	1.2 × 1.2 m	1.44 m²	1.1 m²
F11	Bedroom 1	polygon	15.75 m²	13.87 m²
F12	Owner's office	3.31 × 4.75 m	15.73 m²	13.96 m²
F13	Gym	polygon	22.2 m²	19.92 m²
F14	Suite court (open to the sky)	polygon	25.3 m²	–
F15	Suite lobby	1.35 × 5.4 m	7.29 m²	5.98 m²
F16	Suite bath	polygon	10.24 m²	8.84 m²
F17	Suite WC (closet)	1.1 × 1.5 m	1.65 m²	1.14 m²
F18	Suite dressing	3.01 × 3.45 m	10.39 m²	9.05 m²
F19	Suite bedroom	polygon	26.03 m²	23.6 m²

318.77 m²
outside footprint, incl. courtyard

30.55 m²
courtyard on axes (4.5 × 6.3 clear)

277.76 m²
ground floor rooms (axes)

220.43 m²
first floor rooms (axes), + suite court 25.3 m²

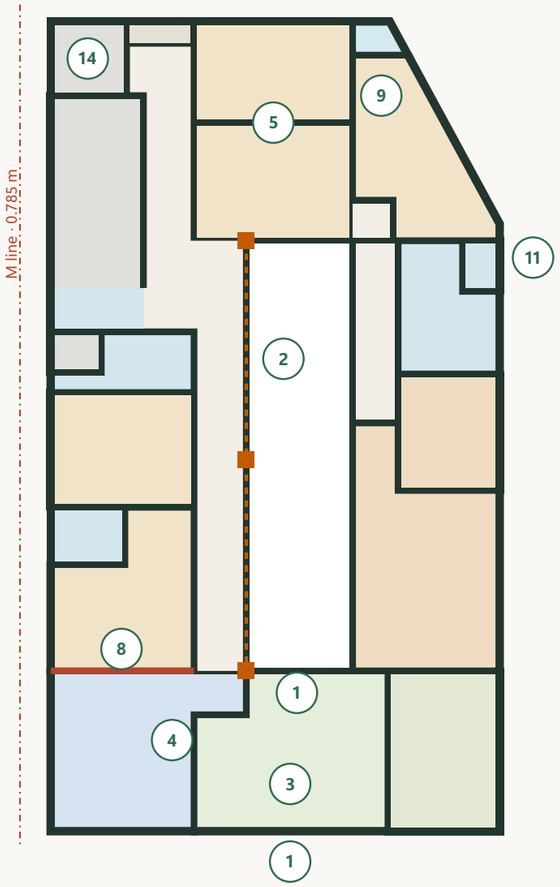
Decisions log

One line per key decision with its reason | from data/decisions.csv | the version is the one in which the decision was made

Ver.	Topic	Decision	Reason	القرار والسبب
V3.0	**Courtyard**	A small courtyard at the centre, enclosed on four sides, open to the sky	The rooms the family uses most look into it; shaded by its walls it stays cool, gives privacy and light from inside	فناء صغير في الوسط مغلق من الجهات الأربع ومفتوح للسماء؛ تظل عليه الغرف الأكثر استعمالاً، ويبقى باردا مظللا ويعطي الخصوصية والضوء من الداخل.
V3.0	**Living room**	One living room with two ceiling heights: double height on the courtyard, normal height (dining) to the east	The double-height half continues the courtyard and works as a chimney; the low half is warmer, easier to cool, and carries the first floor	صالة واحدة بسقفين: نصف بارتفاع مزدوج على الفناء يعمل كمدخنة للتهوية، ونصف بارتفاع عادي للطعام يسهل تكييفه ويحمل الدور الأول.
V3.0	**West service band**	Family entrance, entry cupboard, future lift, stair, utility room with the shaft and the west wash block in one band	Services rise in one place, the structure gains one stiff core, every service is reachable	شريط خدمات غربي واحد: المدخل والصندوق المستقبلي والدرج وغرفة الخدمات مع الشفت وكتلة المغاسل؛ الخدمات في مكان واحد ونواة إنشائية واحدة.
V3.0	**West wash block**	Hospitality kitchen, family basins, family WC, guest WC and guest trough in one strip of wet rooms on one drain run	Short drains, one cleanout, the guest side separated from the family side	كتلة مغاسل ورطبة بخط صرف واحد: مطبخ الضيافة ومغاسل العائلة وغرفة الضيوف والصفيحة الحوضية.
V3.1	**Serving passage**	The only family / guest link is a passage with two doors that are never open together, both locked from the family side	The family serves the men's majlis without being seen; an open guest door shows only a blank wall (majaz)	الصلة الوحيدة بين العائلة والضيوف ممر تقديم بينهما بابان لا يفتحان معا وبقفلان من جهة العائلة؛ يقدم المجلس دون أن يرى العائلة.
V3.3	**Hospitality kitchen**	A light kitchen open to the riwaq for coffee, tea, dates and desserts; the east kitchen is the heavy one	The serving base for the men's majlis sits next to the serving passage, away from the cooking smells	مطبخ ضيافة خفيف مفتوح على الرواق للقهوة والشاي والتمر والحلو، بجوار ممر التقديم والمطبخ الشرقي هو المطبخ الثقيل.
V3.3	**Guest foyer**	A double door at one end of the foyer and the opening into the men's majlis at the other (a bent entry)	Guests do not look straight into the majlis; the majaz is partial in a short foyer (open item)	بهو الضيوف بباب مزدوج في طرف وفتحة المجلس في الطرف الآخر (مدخل منكسر)؛ المجاز جزئي لقصر البهو.
V3.3	**Guest basins**	One trough basin with three taps on the east wall of the guest side	Several guests wash at once before and after the meal	مغسلة حوضية واحدة بثلاثة صنابير في جهة الضيوف؛ يغسل عدة ضيوف معا.
V3.1	**Men's majlis**	A full-width men's majlis on the south, entered only from the foyer; high south windows, seating at the east end	Privacy for the family and the guests; light without being seen	مجلس رجال بعرض المبنى جنوبا لا يدخل إلا من البهو، بنوافذ عالية جنوبية وجلسة رئيسية في الطرف الشرقي.
V3.1	**East service block**	Kitchen on the courtyard lines; laundry on the solid wall of the women's majlis; maid room and maid bath in the sloped corner	The kitchen stays isolated; the maid wing is reached only through the kitchen and the laundry	الكتلة الشرقية: المطبخ على خطوط الفناء، والغسيل على جدار مجلس النساء المصمت؛ وغرفة الخادمة وحمامها في الركن المائل، والوصول للخادمة من المطبخ فقط.
V4.0	**Maid room**	The maid room stays at 7.28 m² clear; the 6.5 m² target is dropped	The sloped cut wall makes the north part too narrow; the freed space went to the maid bath	غرفة الخادمة نحو 7.28 م²؛ صافي وسطها هدف 6.5 م² لأن الجدار المائل يضيق الحيز الشمالي.
V3.4	**Air tower and tunnel**	An air tower in the tip of the cut corner catches the north-west wind above the roof and feeds a tunnel to the courtyard	Cool, clean air without fans; it takes no usable room space; every opening can be closed	برج هواء في زاوي الركن المقطوع يلتقط الرياح الشمالية الغربية ويمررها في نفق إلى الفناء؛ دون أن يأخذ من مساحة الغرف.
V3.3	**Future lift**	A lift shaft built now as a store, fitted later with a two-stop lift (ground and first floor)	The house is ready for a lift now; the roof is reached by the stair	بئر مصعد يبنى الآن ويستعمل مخزنا، وتركب أحقا مصعد بوقفين (أرضي وأول).
V3.0	**Utility room and shaft**	The west service shaft sits inside a utility room with the electrical panel, the water manifold and the water heater	Every service is behind one door and can be maintained without breaking a wall	الشقت الغربي داخل غرفة خدمات مع اللوحة الكهربائية ومجمع المياه والسخان؛ كل الخدمات خلف باب واحد.
V5.0	**Plumbing**	West: the shaft and one drain run under the wash block; east: kitchen, laundry and maid bath straight out east; the suite bath to a stack outside the east wall; no east duct	Short runs, wet rooms stacked over wet rooms, nothing in a kitchen ceiling or over the master bedroom	المصرف: غربا عبر الشفت وحمام واحد تحت كتلة المغاسل، ومتزلة مباشرة إلى الخارج؛ شرقا المطبخ والغسيل وحمام الخدم إلى منصرة خارج الجدار الشرقي؛ لا شيء في سقف مطبخ أو فوق غرفة النوم الرئيسية.
V5.0	**First-floor zoning**	Family part on the north and west around the courtyard; the L (office, gym, suite court, master suite) to the south and east	Children near the shared bath and the family living; the master suite private behind a lobby	توزيع الدور الأول: جزء العائلة شمالا وغربا حول الفناء، وجزء L جنوبا وشرقا (المكتب والنادي وفناء الجناح وجناح الوالدين).
V6.1	**West projection on columns**	The first-floor west band sits on the full 1.55 m projection over the courtyard and the double-height void, on an edge beam and three columns at x 5.8 (each in an existing wall or glass line, a colonnade on the ground floor); the split projection of V5.1 is reverted	A wall on cantilever beams along about 14.95 m costs more and is less direct structurally than walls stacked over walls, and the outward projection 0.11 m from the M line needed municipal approval; rooms and gallery keep their width, the courtyard sky is 3.15 m	الشريط الغربي للدور الأول على بروز كامل 1.55 م فوق الفناء والصالة، على جسر حافة وثلاثة أعمدة عند x 5.8 (في خطوط جدران أو زجاج قائمة (رواق بأعمدة في الأرضي)؛ الفي البروز المقسوم لأن الجدار على كوابيل يطول نحو 14.95 م أعلى وأقل مباشرة من جدار فوق جدار، ولأن البروز الخارجي على بعد 0.11 م من خط M يحتاج موافقة البلدية.
V5.0	**Kids' room**	A kids' room for study and gaming on the north side	Homework and play in one room, away from the bedrooms; it can become a fourth bedroom	غرفة أطفال للدراسة والألعاب في الشمال، يمكن أن تصبح غرفة نوم رابعة.
V5.0	**Family living**	The family living on the courtyard edge is a view, not a path	Nobody walks between the seats and the courtyard	جلسة العائلة على حافة الفناء، أمامك لا ممر.
V5.0	**Bedrooms**	Bedroom 1 one room without a bath; bedroom 2 with a wardrobe niche; bedroom 3 with its own bath over the ground-floor family WC; one family bath by the west shaft	Wet rooms stay on the shafts; every bedroom door opens from the gallery	غرف النوم ٢ و٣ بخزينة، والنوم ٤ بخزينة، والنوم ١ بلا حمام؛ حمام عائلي واحد بالقرب من الشفت.
V5.0	**Master suite**	Lobby, luxury bath with tub, walk-in shower, two basins and a WC closet, walk-in dressing and bedroom, entered from the family link	Privacy (the bed is not seen from the suite door) and the owner's programme	جناح الوالدين: ردهة، وحمام فاخر وغرفة ملابس وغرفة نوم، يدخل منتصر من وصلة العائلة.
V5.0	**Owner's office**	The owner's office in the south-west corner, door from the gym route	A quiet room away from the bedrooms, near the stair	مكتب المالك في أقصى الجنوب الغربي بباب من ممر النادي.
V6.2	**L-shaped gym**	L-shaped gym: the long arm along the south facade (kettlebells, barbell, pilates mat, pull-up bar, one cardio machine) and a short arm at the east end turning north along the suite bedroom wall, holding only the treadmill (0.5 m clear each side, 2 m free behind in the long arm); the suite court becomes about 25.3 m²	The owner's choice: the treadmill gets its own arm. Acoustic separation is mandatory: a double wall to the suite bedroom (about 55 dB), a floating floor or dense rubber under the arm, anti-vibration pads, the treadmill 0.1 m off the wall	نادٍ على شكل L (اختيار المالك): الذراع الطويلة على الواجهة الجنوبية للأثقال وبساط اليوغا، وذراع قصيرة في الطرف الشرقي بمحاذاة جدار غرفة النوم الرئيسية لجهاز المشي فقط؛ مع عازل صوتي: جدار مزدوج (نحو 55 ديسبل) وأرضية عائمة أو مطاط كثيف ووسائد ضد الاهتزاز؛ ويصبح فناء الجناح نحو 25.3 م².
V5.0	**Suite court and clerestory light**	A private court open to the sky in front of the double-height living's clerestory windows; a railing, not a parapet, on the gym roof edge	With a railing about 100 % of the winter noon sun reaches the windows (a parapet would block it)	فناء الجناح المفتوح للسماء أمام النوافذ العلوية للصالة ذات الارتفاع المزدوج؛ ودرابزين لا درابزين؛ على حافة سطح النادي (يصل نحو 100 % من ضوء الشتاء).
V6.2	**Two-storey colonnade (X8)**	The three colonnade columns on x 5.6 continue to the roof inside the gallery railing line; the gallery strip roof rests on them through a roof edge beam (no 1.35 m cantilever); slender round columns, about 0.30 m assumed	Cleaner structure than a cantilevered roof strip; the colonnade reads through both storeys	الأعمدة الثلاثة على x 5.6 تستمر إلى السقف داخل خط الدرابزين؛ سقف شريط الممر يرتكز عليها بكمرة حافة؛ بلا كابولي 1.35 م؛ أعمدة دائرية نحيلة نحو 0.30 م (افتراضي). السبب: إنشاء أنظف.
V6.2	**Drop beams and gypsum ceilings**	Beam B1 (women's majlis / kids' wall) is a drop beam hidden by a gypsum ceiling at about +3.05; rule: drop beams in finished rooms are hidden by gypsum ceilings that also carry the hidden AC and lighting	A flat ceiling in every finished room without special slabs	الكمرة B1 ساقطة وتخفيها سقف جبس عند نحو +3.05؛ القاعدة: الكمرات الساقطة في الغرف المنتهية تخفي بأسقف جبس تحمل التكييف والإضاءة. السبب: سقف مستوٍ في كل غرفة.
V6.2	**Projection edge beam**	Option (b): the edge beam on x 5.6 upturned above the first-floor slab as the base of the gallery railing; (a) a drop beam is acceptable if the engineer decides	A flush colonnade soffit seen from the courtyard; the upstand is the railing base	الخيار (ب): الكمرة الطرفية على x 5.6 مرفوعة فوق بلاطة الدور الأول قاعدة للدرابزين؛ الخيار (أ) الساقطة مقبول إن قرره المهندس. السبب: سقف رواق مستوٍ.
V6.2	**Kitchen service door**	A lockable 0.90 m service door in the kitchen's east wall at its north end near the laundry (y 6.7–7.6) to the east service yard; the cooking counter starts 0.1 m south of it, the east-wall stack moves between the door and the kitchen window	Rubbish, deliveries and gas reach the yard without crossing the house	باب خدمة مقفل 0.90 م في الجدار الشرقي للمطبخ عند طرفه الشمالي قرب المغسل (y 6.7–7.6) إلى فناء الخدمة الشرقي؛ يبدأ رف الطبخ على بعد 0.1 م جنوبه، وتنتقل الماسورة بين الباب والنافذة. السبب: الخدمة دون عبور البيت.
V6.2	**Air system simplified**	Dry malqaf for night ventilation (screen, filter slot, damper); a small dry sealed duct sunlit under the maid wing to a low outlet grille in the courtyard's east wall (the kitchen's west wall) over the sunken seating's bench; the main ventilation by a calculated system (ERV, exhausts, AC) by the MEP engineer	No water in a dark underground duct; the closed rooms need filtered ducted supply; the outlet moved from the north wall to the east wall (3d-chart); the cool air settles where people sit, the women's majlis keeps its full glass	تبسيط نظام الهواء: ملقف جاف للتهوية الليلية (شبكة، فلتر، بوابة إغلاق)، قناة صغيرة جافة محكمة جزيا تحت جناح الخادمة إلى مخرج شبكي في جدار الفناء الشرقي (جدار المطبخ الغربي) فوق مقعد الجلسة المغارة؛ التهوية الرئيسية بنظام محسوب (مبسعه مهندس الكهروميكانيك). السبب: لا ماء في قناة مظلمة؛ يحتاج الهواء المبرد حيث يجلس الناس وبيقى زجاج مجلس النساء كاملا.
V6.3	**TV wall**	The living's west wall (x 4.25) is solid and full height: a 1.40 m arched entrance from the riwaq at its north end, then an arched built-in TV niche 4.40 × 0.70 m (TV 85" centred, low cabinet, side shelves, hidden LED); the sofa faces west, eye-to-screen ≈ 3.45 m, the dining behind; no arches between the colonnade columns	The TV and the sound system need a solid wall; the riwaq no longer looks into the living, so the serving door 1 faces a blank wall	جدار الصالة الغربي مصمت كامل الارتفاع: فتحة مقوسة 1.40 م للمدخل من الرواق في طرفه الشمالي تركد كوة تلفاز مقوسة 4.40 × 0.70 م (للتلفاز 85 بوصة في الوسط، خزانة منخفضة، رفوف جانبية)؛ الكنبة تواجه الغرب بمسافة نحو 3.45 م إلى الشاشة والطعام خلفها؛ لا أقواس بين أعمدة الرواق. السبب: التلفاز ونظام الصوت يحتاجان جدارا مصمتا ولأن مجلس النساء لم يعد مفتوحا على الرواق، وباب الخدمة (القادم) أرصى (أول) قبل السلّم ولا يلغي مجرا مستقلبا؛ أبواب متوالفة من النهو المغاني ومن الممر الغربي؛ انفي خزانة المدخل 0.9–0 م ويصبح الشريط أمام باب المصعد جزءا من البهو؛ مقاس
V6.3	**Lift at construction**	The lift (two stops, ground and first floor) is installed before the family moves in, no longer a future store; sliding landing doors from the family lobby and the gallery; the entry cupboard keeps y 0–0.9 and the strip in front of the lift door is lobby; the shaft 2.25 × 2.2 stays a placeholder until a supplier's civil drawing is chosen	Ageing in place is very important and every bedroom and shower is on the first floor	البئر مؤقت حالي اختيار رسم مورد. السبب: التقدم في العمر مهم جدا وكل غرف النوم والدوشات في الدور الأول.
V6.3	**Door widths**	Every bedroom and bathroom door shuts to a 1.00 m opening with a 0.90 m leaf (≥ 0.85 m clear); two cannot: the guest WC takes a 0.90 m opening (the room is 1.3 m wide) and the maid bath keeps 0.70 m (column C31 and the cut wall) until the maid wing is redesigned; the suite WC closet door opens out into the bath; bedroom 3's door moves to y 15.2–16.2 and its bath door is sliding	Wheelchairs, walkers and furniture pass; a stated requirement of ageing in place	جميع أبواب الغرف والحمامات في الدورين تفتح على 1.00 م (مصفق 0.90 م، عرض الغرفة 1.3 م) وحمام الخادمة يبقى 0.70 م حتى إعادة تصميم جناح الخادمة؛ باب مقصورة مرحاض الجناح يفتح إلى الحمام. السبب: مرور الكراسي المتحركة والمشايات والأثاث.
V6.3	**Suite bath window (X3)**	The suite bath's east window is 0.55 m wide at y 9.50–10.05, so its frame clears column C33 (assumed 0.25 × 0.60 along the wall, y 10.15–10.75) by 0.05 m; the window stays over the tub	No clash with the column whatever size the engineer chooses up to 0.60 m along the wall	نافذة حمام الجناح الشرقية بعرض 0.55 م عند y 9.50–10.05، بحيث يتجه لمرورها عن العمود C33 (0.60 × 0.25 م بطول الجدار) بمقدار 0.05 م. السبب: لا تعارض مع العمود مهما كان مقاسه حتى 0.60 م.
V6.3	**House 0.30 m south (provisional)**	The whole house moves 0.30 m south on the site (shift 3.525 → 3.825); the nearest outer wall face is now 0.20 m from the lane setback line (it governs; the north street line is 0.80 m away) instead of 0.05 m; provisional until the official survey; the south garden is 0.30 m shallower (7.2 m at the middle of the south facade), the palms 5.99 / 7.84 m from the walls	A survey error of a few centimetres would have put the wall in the setback; to reach 0.35 m on the lane the house needs 0.63 m south, or 0.30 m south and 0.17 m west	يُزاح البيت كله 0.30 م جنوبا في الموقع؛ أقرب وجه جدار يبعد الآن 0.20 م عن خط ارتداد الزقاق (وهو المحكم؛ خط الشارع الشمالي على بعد 0.80 م) بدلا 0.05 م؛ مؤقت حتى الرفع المساحي الرسمي؛ الحديقة الجنوبية أقل عمقا بـ 0.30 م والنخلتان على 5.99 / 7.84 م. السبب: خطأ مساحي بسنتيمترات كان سيضع الجدار في الارتداد.
V6.4	**Maid wing to the municipal minimum**	The kitchen's north wall moves 0.50 m south to y 7.0 as one straight non-load-bearing partition; the strip goes to the maid room (≈ 7.28 m² clear, 2.49 m clear at the bed) and the laundry (≈ 6.2 m² clear); the kitchen keeps two parallel runs (≈ 24.0 m² clear); the service door (y 7.2–8.1), the cooking counter (from y 8.2) and the hood (y 10.2–11.2) move with the wall; the y 6.5 line keeps beam B6 as a drop beam in the ceiling under the first-floor wall; the malqaf grille stays at y 7.7–8.7	The municipal rule for a domestic worker's room (≥ 6.5 m² clear, ≥ 2.1 m wide; design review G-04); the structural grid and the first-floor walls are untouched	يتراجع جدار المطبخ الشمالي 0.50 م جنوبا إلى y 7.0 كقاطع غير حامل؛ تكبر غرفة الخادمة إلى نحو 7.28 م² صافي بعرض 2.49 م عند السرير والمغسلة إلى نحو 6.2 م²؛ يبقى المطبخ بصفين متوازيين (نحو 24 م² صافي)؛ يتنقل باب الخدمة وسطح الطبخ والشفاط مع الجدار. السبب: اشتراط البلدية لغرفة العاملة المنزلية (6.5 م² و2.1 م على الأقل).
V6.4	**Courtyard enclosure (option A)**	The conditioned house is closed around the courtyard, which stays open to the sky: the west projection widens 0.20 m to 1.55 m (B3, R11 and C21–C23 at x 5.80, spans and X8 unchanged); black steel divided-lite windows W1 on the gallery edge (on the B3 upstand, guarding transom +4.70, laminated glass below, non-inward casements), W2 on the family living's edge (two casements) and W3 on the living's upper courtyard side at y 13.0 (two motorised tilt lites), heads at +6.75	A closed, filtered house (R-25; design review Y-01, critical) with the gallery ≥ 1.20 m clear (1.235 m at the columns, 1.405 m elsewhere) and the sky opening 3.15 m wide; one line for the beam, the columns and the window (studies/courtyard_enclosure.typ)	يُغلق البيت المكيف حول الفناء الذي يبقى مفتوحا للسماء: يتسع البروز الغربي 0.20 م إلى 1.55 م (الجسر والأعمدة على بعد 5.80 م) وتُغلق الجوانب الثلاث بنوافذ فولاذية سوداء مقسّمة: نافذة المر العلوي على رفع الجسر، ونافذة حافة المعيشة العائلية، ونافذة الجزء العلوي من جهة الفناء للصالة عند 13.0 م، والرؤوس عند +6.75. السبب: بيت مغلق ومفلتر مع ممر علوي لا يظل من 1.20 م صافي وفتحة سماء 3.15 م.
V6.4	**Roof access**	The main stair continues from the first floor to the roof in the same well with the same geometry (two more flights, 11 risers each, trimmers R16–R18 as B16–B18) into a stair room on the well (2.525 × 5.50 inside, 2.20 m clear, roof +9.62) with a door to the roof; the lift stops at the first floor (its headroom 3.40 m to the roof slab underside); the roof is for maintenance and equipment only	Review F-07: no roof access existed; the roof carries the plant, the malqaf hatch, the tank and the PV, and must be reached safely without a ladder; nothing on the first floor is lost	يستمر الدرج الرئيسي من الدور الأول إلى السطح في البئر نفسه وبهندسته نفسها (ثلاثة مداميك إضافية) إلى غرفة درج فوق البئر (2.525 × 5.50 داخليا، ارتفاع صافي 2.20 م) بباب إلى السطح؛ المصعد يقف في الدور الأول؛ السطح للصيانة والمعدات فقط. السبب: لم يكن للسطح منفذ، وعليه المعدات والخزان والطاقة الشمسية.
V6.4	**Roofline (attic and cornice)**	Flat roof, no tiles. A main honey-stone cornice 0.45 m high projecting 0.30 m on the facade edge at roof level all round; on the north block a cream-plastered attic screen 1.8 m high (coping +9.10) set back 1.0 m behind the cornice (its 0.30 m coped upstand at +7.65), hiding the stair room and the roof equipment from the street and the lane at the boundary (the stair room shows 0.6 m above the attic only from more than = 15 m); the south wing keeps the cornice and a plain parapet to +8.40, a 0.70 m step in the roofline; a thin stone band between the floors; no slab level changes	The classical roofline of the design language kept simple; the roof equipment and the stair room hidden from the public eye; the courtyard loses no sky or sun (the attic lies below the sky hole's own edges from every point of the courtyard floor)	سطح مستوٍ بلا قرميد؛ كورنيش رئيسي من الحجر العسلي على حافة الواجهة عند السطح حول الكتلة الشمالية مآجر سطح (عُلّية) بارتفاع 1.8 م مرتدّ 1.0 م خلف الكورنيش يخفي غرفة الدرج والمعدات؛ المآجر لا تظهر من الشارع والزقاق إلا من مسافة 15 م فأكثر؛ الجناح الجنوبي كورنيش وشرفة عادية على ارتفاع 8.40 م، مع درجة 0.70 م في خط السطح؛ حزام حجري رفيع بين الدورين. السبب: لغة التصميم الكلاسيكية البسيطة وإخفاء المعدات.
V7.1	**Arched enclosure windows**	The courtyard enclosure windows get semicircular (Roman) heads in black steel with divided lites, the arched heads fixed and the operable lites below: W1 six arched windows 1.60 m wide on the gallery edge (three per column bay, over the arcade bays; sill +3.82, springing +5.80, crown +6.60), W2 two arched windows 1.30 m east of C21 (floor to +6.60), W3 one arched window 2.80 m over the living's courtyard glass (+3.32 to +6.60, springing +5.20, the tilt lites below); every crown 0.15 m below the roof edge beams' soffit	The design language (Roman arches, black steel divided lites); the courtyard reads as a two-tier classical arcade with the arcade below	تأخذ نوافذ إغلاق الفناء رؤوسا نصف دائرية (رومانية) من الفولاذ الأسود المقسّم: ست نوافذ مقوسة عرض 1.60 م على حافة الممر العلوي فوق أقواس الرواق، ونافذتان 1.30 م على حافة المعيشة العائلية، ونافذة كبيرة واحدة 2.80 م فوق زجاج الصالة؛ كل تاج 0.15 م تحت جسور حافة السطح.
V7.1	**Courtyard arcade**	In the courtyard bay only (between C21 and C22) two slender honey-stone columns Ø0.25 on x 5.80 at y 8.667 and 10.833 give three equal bays of 2.167 m with three semicircular arches under the projection (spans 1.84 / 1.92 / 1.84 m, springing +2.31 … +2.35, crown at the soffit +3.27); decorative unless the engineer uses the columns; the W1 arched windows above align with the three bays; the living bay keeps no arches	A two-tier classical arcade round the courtyard (the design language); the columns stand clear of the sunken seating (0.78 m), the riwaq	في جهة الفناء فقط عمودان رشيقان من الحجر العسلي قطر 0.25 م على خط الرواق 5.80 م يقسمانه إلى ثلاث فتحات متساوية بثلاثة أقواس نصف دائرية تحت البروز (زخرفية ما لم يستعملها المهندس)؛ النوافذ المقوسة فوقها على محاور الأقواس؛ جهة الصالة بلا أقواس.
V8.0	**Fire safety: early warning and safe exit (home level)**	Home level: interconnected mains smoke alarms with battery back-up in every bedroom and on the routes, a heat alarm in the kitchen, CO and gas detectors, extinguishers and a fire blanket; escape casements without screens in every sleeping room (the maid room's at sill +1.10); exit doors with thumb-turns. Not adopted: sprinklers, fire doors, a panel, a safe room	Owner; studies/fire_escape.typ	على مستوى المنزل: إنذار مبكر مترابط ونوافذ هروب بلا شبك وأبواب خروج بلا مفتاح؛ بلا رشاشات أو أبواب حريق.
V8.0	**Courtyard layout**	The seating one step down (−0.15) from the colonnade edge to the kitchen wall with three benches and a low table; the outdoor TV in a semicircular stone surround on the kitchen wall, the malqaf grille below it; the north enclosure with a water basin and one tree; no pergola; the drainage section	Owner; studies/courtyard_layout.typ	جلسة منخفضة درجة واحدة بمصاطب وطاولة، وتلقاز في إطار حجري، وحوض ماء وشجرة؛ بلا عريشة.
V8.0	**Kitchen sink window moved north**	Moved north to y 7.3–8.9 (1.6 m, sill +1.10) over the sink, the dishwasher south of it; the outdoor TV, its surround and the malqaf grille stay (0.65 m of wall to the frame)	It looks at the fountain and the tree; the TV wall stays solid	تنتقل نافذة الحوض شمالاً إلى y 7.3–8.9 (1.6 م، سل +1.10) فوق الحوض، والغسالة جنوبه؛ التلفاز وإطاره والمشربية تبقى في مكانها.
V8.0	**Air system v3 (owner goals; proposals assumed, MEP to design)**	One tower, two separate paths: a gentle courtyard breeze in pleasant weather (not cooling) and the fresh-air intake of the ventilation unit; no water, pads, earth pipes or human entry (v1 / v2 withdrawn). Proposals: a four-sided head with flaps, a split shaft, a spring-return damper, a 0.9 × 0.20 channel in the ground slab with covers, the grille under the TV, a 750 m³/h HRV in a roof plant room; an assist fan proposed, not added	Review E-01: a sealed house needs a real ventilation unit; the malqaf stays the symbol and the breeze	برج واحد بمسارين منفصلين: نسيم للجلسة ومدخل الهواء النقي لوحدة التهوية؛ لا ماء ولا دخول بشري. المقترحات اقترا ضية.
V8.0	**Air system v3.1: open items decided**	Assist fan YES (a quiet EC extract-type fan in the plenum behind the grille, own switch by the seating, slid out through the grille for service); bird mesh only on the breeze path; the HRV in a base under the malqaf tower, one mass, top +9.10 (nothing but the tower above the attic), serviced from the roof through doors; the AC condensers south of the tower (2.2 m of the base); the kitchen make-up inlet discreet under the stone band, interlocked with the hood	Owner's decisions on the study's open items (studies/air_system.typ v3.1)	مروحة مساعدة هادئة خلف القحة بمفاتحها؛ شبك طيور فقط؛ وحدة التهوية في قاعدة تحت برج الملقف ككتلة واحدة؛ المكيفات جنوب البرج؛ مدخل هواء المطبخ مخفي تحت الحزام الحجري.
V8.0	**South wing swapped (applied)**	SUPERSEDED by south wing v2: the gym on L on the west and the south with a 2.5 m court and the office east	Owner; replaced the same day	استُبدل بالنسخة الثانية من الجناح الجنوبي.
V8.0	**External stair tower east; roof stair dropped; outdoor kitchen**	An enclosed arched stair tower outside the east wall at the south-east (2.6 × 5.78, top +9.60) to the office landing and the roof, a locked gate above the landing, an out-opening gate at the foot, no door into the house, a service gate in the lane wall; the internal stair stops at the first floor; the outdoor counter at the tower's east face, the grill 3.4 m south of the gate	Feasible with conditions (suite window shortened to y 15.0–17.9); counted as the first floor's second way out	برج درج خارجي في الزاوية الجنوبية الشرقية إلى المكتب والسطح؛ الدرج الداخلي يتوقف عند الطابق الأول؛ المطبخ الخارجي عند البرج والشواية بعيدا.
V8.0	**South wing v2**	Gym on the west band (≈ 18 m² clear, treadmill on the 2.75 m solid bay, an entrance nib at the gallery end); the suite court from the gym to the office (≈ 25 m², the three doors on a 1.2 m strip, seating south); the office east, no door to the bedroom; fixed TVs; R6 removed	Owner; X5 improved, not verified	الصالة غربا؛ فناء الجناح الكبير من الصالة إلى المكتب؛ المكتب شرقا بلا باب إلى غرفة النوم.
V8.1	**Stair tower removed; roof by a maintenance ladder; roofline lowered**	The external stair tower, its gates, landings and the visitors' door to the office are removed; office visitors are met in the men's majlis, the office is reached from the suite court only (simple doors, no visitor locks); the roof by a manufactured vertical ladder in a 0.90 × 0.30 recess in a honey-stone pilaster on the gym's east wall in the suite court (3.7 m clear), behind a locked walnut-finish panel, a self-closing gate in the court guard at the top, a central fall-arrest rail, the stiles 1.10 above the roof; the attic lowered to 1.40 m above the roof finish (coping +8.70), the south wing parapet 0.70 m (+8.00) with a continuous 1.10 m black steel guardrail set back 1.0 m (the strip outside it a no-go zone); the outdoor counter at the east wall, the grill apart; the service gate on the lane proposed to keep for the yard	Owner, 1 Nov 2026 after V8.0 (cost and appearance); the sight lines re-run: 1.40 hides 1.2 m condensers on 0.10 pads and the tank (studies/roof.typ § 3); the HRV base (+9.10) stays 0.40 above the attic as part of the malqaf mass, or is lowered (option)	إلغاء برج الدرج الخارجي وباب الزوار؛ المكتب من فناء الجناح فقط؛ الوصول إلى السطح بسلم صيانة ثابت في فناء الجناح خلف لوح مقلّ؛ خفض العلية إلى 1.40 م والدروة الجنوبية إلى 0.70 م مع درابزين مرتكز 1.10 م.
V8.2	**Main stair flipped (option B) and the family bath enlarged (F-b)**	The main stair turned round in the same well: the lower flight on the east half rises south straight off the family lobby (the start at the north end, 14.5 m in front of the first riser), the half landing at the south end, the upper flight on the west half arrives at the north end beside the lift's first-floor door; the trimmers B16'–B18' mirrored (headroom 2.22 m at tread 6, 0.30 trimmers an option for 2.32); a 0.90 black-bar balustrade along the lower flight on the riwaq; the store under the half landing with a low door onto the riwaq; a small obscured west window at the half landing; a glazed obscured side light at the family entrance; the small entry cupboard removed, a full-height coat cupboard in the west half of the start strip. The freed south landing becomes the family bath's north part (an L) with a separate WC compartment by the shaft stack and a double basin	Owner, 2 Nov 2026, from studies/stair_flip.typ: the study room, the living, the suite and the entrance come 5–7 m nearer the stair, bedrooms 2 and 3 lose 3.8 m; a proper foot; the same headroom and structure	قلب الدرج الرئيسي (الخيار ب): البداية والوصول في الطرف الشمالي بجوار المصعد، وبسطة الوصول القديمة تصبح جزءا من حمام العائلة مع مرحاض منفصل ومغسلتين؛ درابزين على الرواق ونافذة صغيرة وزجاج مطلّل بالمدخل.
V8.2	**Food route to the men's majlis accepted (review X-01)**	The food route from the kitchen through the dining and the riwaq to the serving passage is accepted as drawn: no outside service path, no serving door in the majlis	Owner, 2 Nov 2026: normal guest numbers; very large events use the old majlis on the farm; the owner serves	قبول مسار الطعام من المطبخ إلى مجلس الرجال كما هو مرسوم: لا مدخل خدمة خارجيا، ولا باب تقديم في المجلس.
V8.2	**Bedroom 1: no private bathroom (option A; review F-05)**	Bedroom 1 keeps its size and uses the family bath (two compartments since the stair flip); the wet areas stay grouped in the two existing clusters (west round the shaft, east along the east wall); low-level motion-sensor night lights along the route bedroom 1 → family link → family living → gallery → family bath (design_ff.typ marks; a note for the electrical design)	Owner, 2 Nov 2026, from studies/bedroom1_bath.typ: the 17.3 m route accepted; no third wet cluster	غرفة النوم ١ بلا حمام خاص (الخيار أ): تبقى بحجمها وتستخدم حمام العائلة، مع إضاءة ليلية منخفضة بحساس حركة على المسار.
V8.2	**Ground-floor shower deferred to the plumbing study (review G-07)**	Deferred: decided with the plumbing study (open item 10); P1 (the family WC and the family basins merged into one accessible wet room with a level shower and a 1.5 m turning circle) stays the preferred option, P1-lite (the family WC room only) the fallback	Owner, 2 Nov 2026, from studies/gf_shower.typ	دش الدور الأرضي مؤجل إلى دراسة السباكة؛ الخيار المفضل دمج دورة العائلة والمغاسل في حمام مبيّن، والبديل دورة العائلة وحدها.
V8.2	**Suite court south side: a mashrabiya screen**	The court's south side is a mashrabiya screen on the 1.10 m base up to the facade frame's soffit (the full height of the opening under the parapet), timber or black steel lattice in the design language: the palms seen out, no view in	Owner, 2 Nov 2026 (studies/south_wing.typ § 12 option b)	الجهة الجنوبية لفناء الجناح: مشربية بكامل ارتفاع الفتحة فوق القاعدة 1.10 م بلغة التصميم؛ ترى النخيل ولا يُرى الداخل.
V8.2	**Clerestory windows stay openable (open item 14 closed)**	The double-height living's clerestory windows stay openable vents (ventilation, the living's smoke outlets); the suite court is no longer an escape route since the stair tower went, so fixed fire-rated glazing is not needed; privacy by the 1.40 m sill and the court's mashrabiya	Owner, 2 Nov 2026	النوافذ العلوية للصالة تبقى قابلة للفتح؛ لا حاجة لزجاج مقاوم للحريق لأن فناء الجناح لم يعد مخرج هروب.
V8.2	**Site plan: the north-strip parking kept; the inverter and batteries in the utility room**	The uncovered family parking space north of the house (5 × 2.5 in the setback strip, 0.4 m off the wall, clear of the family entrance) is KEPT; a 3-point manoeuvre from the driveway (studies/site_plan.typ). The future inverter and batteries go in the utility room (west), away from the water: the heater and the manifold on the room's south side, the batteries on the north wall under the family bath's dry linen zone (the bath's slab sunken and tanked); the electrical engineer confirms the space, the ventilation and the fire safety	Owner, 2 Nov 2026 (the municipality allows parking inside the plot)	موقف السيارة شمال البيت يبقى (مناورة بثلاث حركات)؛ العاكس والبطاريات في غرفة الخدمات غربا بعيدا عن الماء، ويؤكد مهندس الكهرباء المساحة والتهوية والسلامة.
V8.3	**Bedroom 1: en-suite shower room, hotel layout, B4 · studied, NOT applied**	B4 (an entry zone by the link, two doors side by side on one line, a 1.6 × 2.4 shower room over the laundry / maid bath, a ≥ 10 m² clean bedroom) fails its acceptance criteria 2 and 3 in every variant: bedroom 1 is a 13.9 m² wedge and the programme needs ≥ 15.7 m²; the best variant B4-b leaves a 7.3 m² L-shaped bedroom with the two doors at a corner (studies/bedroom1_bath.typ page 3). Option A (2 Nov) stands for now; since the family bath is a WC only, bedroom 1 has no shower on its floor until the owner decides (B4-b with its two failed criteria, or another answer)	Owner's brief 3 Nov 2026; the sizes are targets, nothing squeezed	حمام غرفة النوم 1 بتخطيط فندقي (B4): دُرس ولم يُطبق لأن معيارين من معايير القبول لا يتحققان (الغرفة مثلثة الشكل)؛ يقرر المالك بين B4-b مع نقصه أو حل آخر.
V8.3	**Bedroom 2: en-suite bathroom; the first-floor family bath becomes a family WC**	The F-b bath by the west shaft is split: bedroom 2's en-suite (y 9.2–11.0 with the former wardrobe niche as its vanity niche, ≈ 4.7 m² clear: a level shower 1.0 × 1.6 with a channel along the service wall, a wall-hung WC, the vanity; its door from inside bedroom 2) and a family WC (y 7.9–9.2, ≈ 2.8 m² clear, a wall-hung WC behind a screen wall, a basin; its door from the gallery). Every fixture on one 0.20 service wall on y 9.2 into the shaft (no pipe in the floor); the gallery and the guard round the stair opening untouched; bedroom 3's bath unchanged. Result: the suite, bedrooms 2 and 3 have their own bathrooms plus a family WC upstairs; bedroom 1 waits (B4)

Open items / for the owner and the engineers

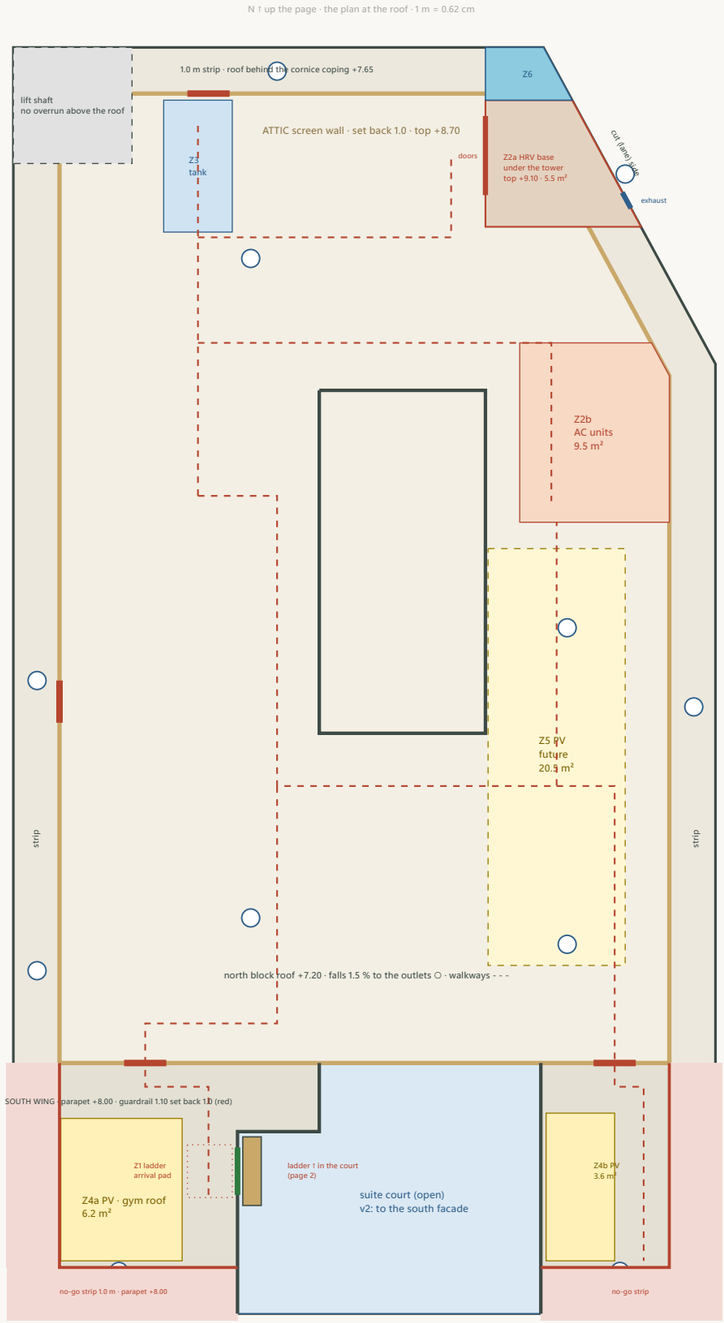
From data/open_items.csv (the same list as docs/OPEN_ITEMS.md) | orange = the edge beam and the 3 columns, red = the acoustic wall gym ↔ bedroom 3 | the roof ladder in the suite court



No.	For	Item	Action	البند
1	Survey	Envelope gap 0.2 m from the nearest outer wall face to the setback line (the lane side; V6.3: the house moved 0.30 m south as a provisional margin)	Confirm by the official survey before the wall faces are fixed	المسافة 0.2 م بين أقرب وجه جدار وخط الارتداد: تؤكد بالرفع المساحي الرسمي.
2	Structural	Edge beam on x 5.8 (spans 6.5 and 6.25 m) and three columns at x 5.8 under the first-floor west projection	Engineer: beam depth, column sizes and foundations; the columns stand in the women's majlis south wall, the courtyard / living glass and the living's south wall	جسر الحافة على x 5.8 (بحور 6.5 and 6.25 م) والأعمدة الثلاثة تحت البروز الغربي: يحدد المهندس الإنشائي المقاسات والأساسات.
3	Structural	South wing v2 over the men's majlis and the guest block (X5 revised, improved, not verified): the gym inside the column frame (B19 between C19 and C20, a new B22 under the double wall), the suite court terrace on the majlis roof, the office wall x 10.0 on rib B21 (non-load-bearing), R8 across the court's south side	Engineer: B19, B22, R8, the terrace load (depressed zone, falls, planters), the treadmill bay's vibration	الجناح الجنوبي فوق مجلس الرجال وكتلة الضيوف: الجسور B19 وB22 وR8 وحمل فناء الجناح واهتزاز جهاز المشي، للتأكيد من المهندس.
4	Structural	The maintenance ladder in the suite court (V8.1): the honey-stone pilaster on the gym's east wall (the 0.20 wall not cut, X20), the ladder's fixings, the gate posts on R3; the 1.10 m guardrail set back 1.0 m on the south wing (anchors through the waterproofing)	Engineer and the ladder supplier; the fall protection confirmed by the specialist under the Saudi requirements	سلم الصيانة في فناء الجناح: العمود الحجري على جدار النادي وتثبيت السلم والبوابة، والدرابزين المرتد على الجناح الجنوبي؛ يؤكد المهندس والمختص.
5	Structural	First-floor walls off ground-floor lines (y 3.0 over the women's majlis, the suite walls over the kitchen and, if the plumbing proposal (b) is adopted, the suite bath's 0.20 service wall on the depressed slab, the gym nib and the office wall over the men's majlis; the family bath's wall on y 7.9 is on B12)	Engineer: light partitions on the slab or beams	جدران الدور الأول خارج خطوط الأرضي: قواطع خفيفة أو جسور.
6	Structural	Floor-to-floor height 3.60 m (assumed); the flipped stair (V8.2): headroom 2.22 m at tread 6 under the mirrored trimmers B16′–B18′ (0.30 trimmers give 2.32), the first-floor opening turned round; handrails	Engineer to confirm the levels	ارتفاع الدور 3.60 م (مفترض)؛ الدرج المعكوس: ارتفاع حر 2.22 م تحت الجسور الحاملة للفتحة؛ يؤكد المهندس.
7	Structural	No soil report: the footing type and depth, the lift pit, the breeze channel's base and sump	Owner: commission the soil investigation	لا يوجد تقرير تربة: نوع القواعد وعمقها وحفرة المصعد وقناة النسيم وقواعد برج الدرج.
8	Acoustics	Acoustic wall between the gym and bedroom 3 (south wing v2): a double wall (target Rw about 55 dB airborne), a floating floor or dense rubber under the treadmill, anti-vibration pads, nothing fixed to the wall	Acoustic / structural engineer: the build-ups, then a test on site	جدار مزدوج بين النادي وغرفة النوم 3 (نحو 55 ديسيبل) وأرضية عائمة تحت جهاز المشي.
9	MEP	Air system v3.1 (AIR_SYSTEM.md): one tower, two separate paths; the HRV (750 m³/h class) in the base under the malqaf, the risers, the ducts, the filters, the courtyard breeze path with its assist fan, the kitchen make-up air; the head's south face sits behind the base (three open faces)	MEP engineer: design and size everything marked assumed	نظام الهواء v3.1: برج واحد بمسارين؛ وحدة التهوية في قاعدة الملقف؛ يصمم مهندس الميكانيكا كل المقاسات.
10	MEP	Plumbing study rounds 1–2 (plumbing.typ, PROPOSALS) with the owner's decisions of 3 Nov 2026 (the east stack external behind cladding, automatic trap primers, rain / sewage separate, condensate by tundishes); still open: the suite bath's drainage approach (a: all in the sunken slab / b: the east service wall + the tub's plinth), the heaters (two cylinders or a roof heat pump), the utility room's door moved south, the water system (the roof tank, the booster, any greywater line), the septic's inlet level (a pump station if too high); every size assumed	Owner: the suite bath approach, the heaters, the utility door; then the plumbing engineer designs the falls, the pipe sizes, the stacks and the pump station if needed	دراسة السباكة: قرارات 3 نوفمبر مسجلة (الماسورة الشرقية خارجية، مغذيات السيفونات، فصل الأمطار)؛ ما زال مفتوحا: أسلوب صرف حمام الجناح والسخانات وباب غرفة الخدمات ونظام الماء؛ ثم يصمم المهندس.
11	MEP	The stack outside the east wall (suite bath + vent) at y 7.05 behind a removable cladding panel (plumbing round 2, A3: recommended over an internal duct), its four branch fittings at +3.4 … +3.9 and a rodding eye; the kitchen extract through the east wall; the hood clear of the stack	Owner: confirm the external stack; MEP engineer to size and detail it	الماسورة خارج الجدار الشرقي خلف كسوة قابلة للفك (موصى بها على القناة الداخلية)؛ يؤكد المالك ويصمم المهندس.
12	Owner	Service door to the east service yard: DECIDED, a lockable 0.90 m door in the kitchen's east wall at its north end (y 6.7–7.6), near the laundry; drawn	Lock and hardware with the joinery; MEP moves the stack south of it	باب الخدمة: تقرر. باب مقفل 0.90 م في الجدار الشرقي للمطبخ عند طرفه الشمالي قرب الغسيل إلى الفناء الشرقي (نفايات، توصيل، غاز)؛ مرسوم. القفل مع النجارة.
13	Owner	Foyer majaz only partial: the entrance door and the majlis opening overlap	Owner: accept, or add a curtain or a short screen	المجاز في بهو الضيوف جزئي: قبول أو ستارة أو حاجز قصير.
14	Owner	Lift installed at construction: the shaft 2.25 × 2.2 is a placeholder until a supplier's civil drawing is chosen (two stops); the roof slab is closed over the stair well and the lift does not serve the roof	Owner: ask 2 or 3 suppliers, use the largest shaft; the headroom above the first-floor stop ≤ 3.40 m (≤ 3.70 with a 0.30 raised head hidden behind the coping)	المصعد عند الإنشاء: المقاس مؤقت حتى اختيار المورد، وارتفاع الرأس ≥ 3.40 م.
15	Design	Daylight check of the courtyard and the double-height living	A daylight simulation with the final levels	فحص الإضاءة الطبيعية للفناء والصالة بمحاكاة.
16	Design	Bedroom 1 has no shower on its floor since the family bath became a WC (3 Nov 2026); its hotel-layout en-suite B4 failed the acceptance criteria (studies/bedroom1_bath.typ page 3: the wedge room); the best variant B4-b leaves a 7.3 m² L bedroom with the two doors at a corner	Owner: B4-b with its two failed criteria, bedroom 2's or 3's en-suite shared, or another answer	غرفة النوم 1 بلا دش في دورها بعد تحويل حمام العائلة إلى دورة مياه؛ خيار B4 لم يحقق المعايير؛ يقرر المالك.
17	Design	The facade principle R-37 (no privacy from outside: the house inside a fully walled farm) to be applied at the facade design stage: the walls now blank or with small / high windows for privacy (the women's majlis north window, the kids' room, a north-light clerestory for the family living, the dining and suite east windows, the office, the men's majlis sills) are candidates to open; the west guest side keeps its privacy	Designer at the facade stage, with the climate rules (shading, dust) and the escape windows	مبدأ الواجهات (لا خصوصية من الخارج): يطبق في مرحلة تصميم الواجهات بفتح الجدران المغلقة والنوافذ الصغيرة إلا في الجهة الغربية (طريق الضيوف).
18	Survey	The existing septic OUTSIDE the farm wall near corner C1 (its position, inlet level, capacity, condition), the existing ground tank east of peg R, peg R and corner C1 themselves, the street level at the gate, the farm's levels: all approximate in studies/site_plan.typ	Owner: the survey; then the drainage falls are re-checked (a pump station at the gate's collector chamber if the inlet is higher than −1.19)	خزان التحليل خارج السور قرب C1 (موقعه ومنسوب مدخله وسعته وحالته) وخزان الماء شرق الوتد R والمناسيب: بالرفع المساحي؛ ثم تُراجع ميول الصرف.
19	Owner	The municipality: the septic's reuse and its legal position outside the wall (the drain crossing under it), the rain soakaways and their overflows, the gas cabinet and the Civil Defense's villa requirements (access, water, alarms); parking in the setback strip is allowed (owner, 2 Nov 2026)	Owner: ask the municipality and the Civil Defense with the site plan	البلدية: إعادة استعمال خزان التحليل خارج السور، وآبار تصريف الأمطار، وخزانة الغاز ومتطلبات الدفاع المدني.

Roof plan / the roof ladder, the set-back attic, the set-back guardrail and the reserved zones

studies/roof.typ (V8.1: the roof by a maintenance ladder in the suite court; the internal stair stops at the first floor; the malqaf on its HRV base) | flat roof, no tiles | the attic 1.40 m above the finish, set back 1.0 m | the south parapet 0.70 m with a 1.10 m guardrail set back 1.0 m | zones reserved, not designed | all sizes assumed



Access (V8.1): by a **maintenance ladder in the suite court**: a manufactured vertical ladder in a 0.90 × 0.30 recess of a honey-stone pilaster on the gym's east wall, behind a locked walnut-finish panel (not climbable when closed); 3.7 m from the court floor to the gym roof, the stiles 1.10 m above the roof as the transfer handholds (+8.40), a self-closing gate in the court's west guard, a landing pad inside the guarded zone; a fall-arrest rail, to be confirmed by the specialist. The internal stair stops at the first floor; the roof slab over the well is closed; no door from the roof into the house. The lift stops at the first floor: headroom to the roof slab underside 3.4 m (a lift with ≤ 3.40 m, or ≤ 3.70 m with a 0.30 m raised head hidden behind the cornice; open item 15).

Roofline (V8.1, lower and balanced): a honey-stone **main cornice** 0.45 × 0.3 m at +6.90 … +7.35 all round; on the north block a coped upstand to +7.65 and the **attic screen** 1.4 m above the roof finish (coping +8.70) set back 1 m on all four sides; the **south wing** a low parapet 0.7 m above the finish (+8.00) and a **continuous 1.10 m black steel guardrail set back 1 m** (not seen from the street; the strip outside it a no-go zone): the roofline steps 0.7 m. Above the attic only the **malqaf** (+9.80) on its HRV base (+9.10) at the north-east. The attic hides 1.2 m condensers on pads and the tank from every distance (§ 3 of the study). The attic never cuts the courtyard's sky or sun.

Zones (reserved): the HRV in the malqaf's base; the AC units behind the cut / east attic wall; water tank(s) between the roof beams R10 and R3; PV on the gym roof and the office roof inside the guardrail and, later, on the north block's east roof; the 1.0 m strip in front of the attic (a no-go zone) with its own outlets; 11 roof outlets to internal downpipes, falls 1.5 %; 0.9 m walkways from the ladder's landing pad through the attic opening at x 2.1–2.9 to every zone, with a branch to the office roof; the guarding continuous from the ladder to every piece of equipment (checked).

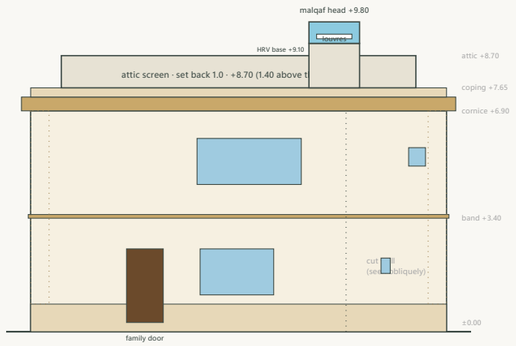
Zone	m²	Note
Z1 the maintenance ladder (suite court → gym roof)	0.5	V8.1 (owner): the roof is reached by a fixed vertical ladder in a recess in a pilaster on the gym's east wall (page 2): 3.7 m climb from the court floor +3.60 to the gym roof +7.30, behind a locked walnut-finish panel; the internal stair stops at the first floor, the roof slab over the well closed
Z2a the HRV base under the tower	5.5	`studies/air_system.typ` v3.1 (owner): the tower rises from a base housing the ventilation unit (one mass): top +9.10 (0.4 m above the lowered attic coping: part of the malqaf mass on the facade, seen by design; § 8), 1.65 m clear inside (a low service compartment: the unit ≈ 110 kg on a plinth right behind a pair of doors 1.5 m high on its west face, filters and the core pulled from the roof terrace, no entry); the intake duct drops inside the tower's compartment; the exhaust louvre on the cut face
Z2b plant: AC outdoor units	9.5	behind the cut / east attic wall, SOUTH of the tower and its base (owner): 2.2 m clear of the base's south wall, 5.15 m south of the intake louvre in plan, so the tower never takes their hot air; over the suite bath (anti-vibration mounts); 0.6 m service space round each unit
Z3 water tank(s)	3.3	≈ 2 × 1.5 m³ between the roof beams R10 and R3, inside the attic (needed only if the owner wants a gravity reserve)
Z4 PV, south wing (gym roof Z4a + office roof Z4b)	9.8	west and east of the court, never in front of the clerestory windows; V8.1 only inside the set-back guardrail and off the walkways (was 24.6 m²: the no-go strips and the ladder's walkway take the rest); fixing points cast in before waterproofing
Z5 PV, future (north block east)	20.5	inside the attic; morning shade from the east attic wall
Z6 malqaf head	1.4	the tower rises to +9.80: with its base the one element above the attic, by design; v3: a four-sided head, the breeze shaft and the ventilation intake compartment inside (`studies/air_system.typ`)
Z7 the 1.0 m strip in front of the attic	40.6	roof behind the main cornice and its coping; falls to its own outlets; a NO-GO zone (the coping is 0.35 above the finish): its two openings in the attic wall get locked gates; entered only on a harness clipped to anchor eyes on the attic wall
Z8 the no-go strips of the south wing	13.6	the 1.0 m between the guardrail and the parapet (+8.00, 0.70 above the finish): no equipment, no walkway; the outlets there reached on a harness

السطح

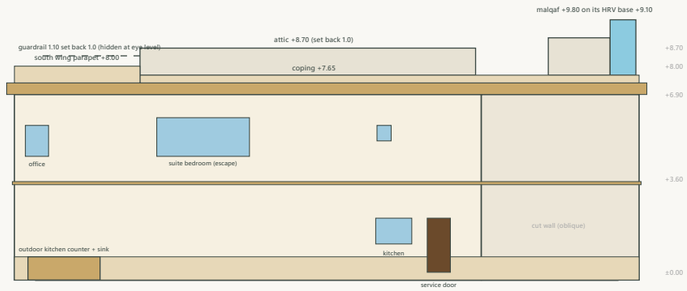
الوصول إلى السطح بسلم صيانة ثابت في فناء الجناح خلف لوح مقفل (V8.1)، والدرج الداخلي يتوقف عند الدور الأول وبلاطة السطح فوق البئر مغلقة؛ المصعد يقف في الدور الأول. كورنيش رئيسي من الحجر العسلي حول البيت، وعلى الكتلة الشمالية حاجز سطح بارتفاع 1.4 م مرتدّ 1 م خلف الكورنيش يخفي المعدات؛ الملقف على قاعدة وحدة التهوية؛ الجناح الجنوبي بدروة منخفضة 0.70 م ودرابزين مرتد 1.10 م فيتدرّج خط السطح 0.7 م. مناطق محجوزة: وحدات التكييف خلف العلية، الخزان، الألواح الشمسية، الممرات والمصارف والحواجز.

Elevations / the lowered roofline and the sight lines

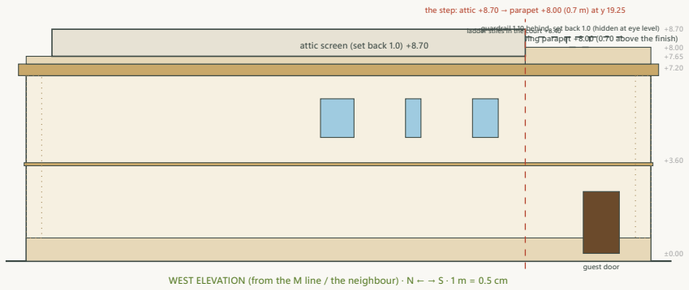
studies/roof.typ (V8.1) | cream plaster, honey stone at the plinth, the corners, the band between the floors and the main cornice | black steel divided-lite windows | the attic set back 1.0 m | openings indicative



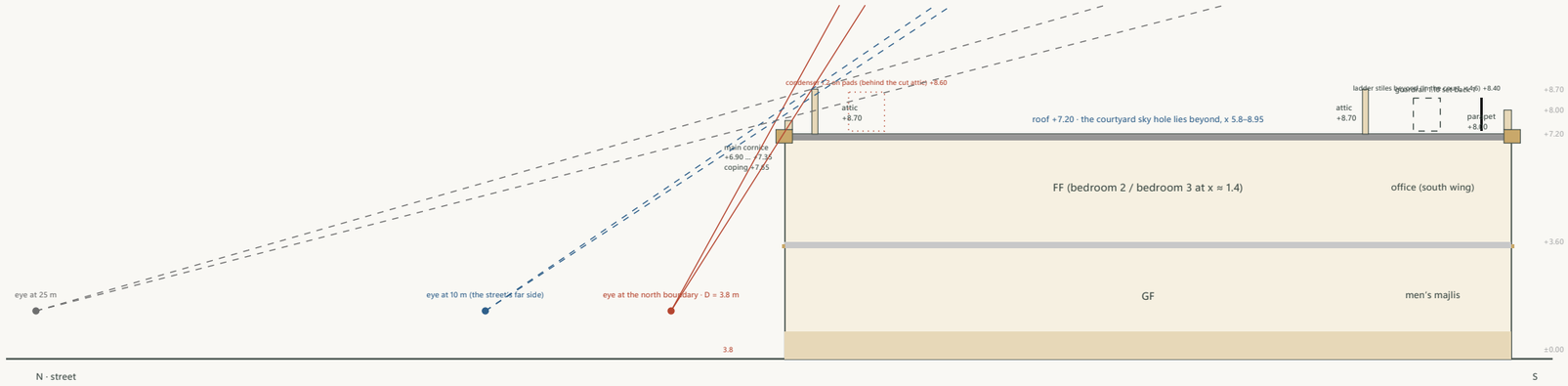
NORTH ELEVATION (from the street) · W ← → E · 1 m = 0.5 cm



EAST ELEVATION (from the lane) · S ← → N · 1 m = 0.42 cm (V8.1: no stair tower)



WEST ELEVATION (from the M line / the neighbour) · N ← → S · 1 m = 0.5 cm



Sight lines (eye 1.6 m above the ground, +1.30 against the house): from the north boundary (D = 3.8 m) the cornice coping (+7.65) is seen at 59.1° and the attic coping at 57°: the attic hidden behind the cornice. Nothing stands above the attic but the malqaf on its base (by design). The attic (V8.1: 1.4 m above the finish) hides 1.2 m condensers on 0.10 pads (+8.60) and the tank from the street and the lane at every distance: the minimum height is 1.3 m; the units are specified ≤ 1.2 m high.

Elevations: the main cornice (0.45 m, projecting 0.3) at +6.90 the thin stone band at +3.40 the plinth to +0.60 and the quoins in honey stone; the north block's coping +7.65 with the attic (+8.70) 1.0 m behind it; the south wing's low parapet +8.00 with the guardrail set back behind it; the malqaf head +9.80 on its HRV base in the cut tip; on the east the office's window and the outdoor kitchen's counter at the south end (V8.1). No element above the attic on the west.

الواجهات وخطوط النظر

من حدّ الأرض الشمالي تختفي العِلّية خلف الكورنيش، ولا شيء فوقها إلا الملقف على قاعدته؛ العلية بارتفاع 1.40 م تخفي المعدات، والدروة الجنوبية 0.70 م مع درابزين مرتد لا يُرى من الشارع.

الواجهات: لياسة كريمية، حجر عسلي في القاعدة والزوايا والشريط بين الدورين والكورنيش الرئيسي، نوافذ فولاذية سوداء مقسّمة.